

Data Transfer Object (DTO) Design Pattern

Lecture notes

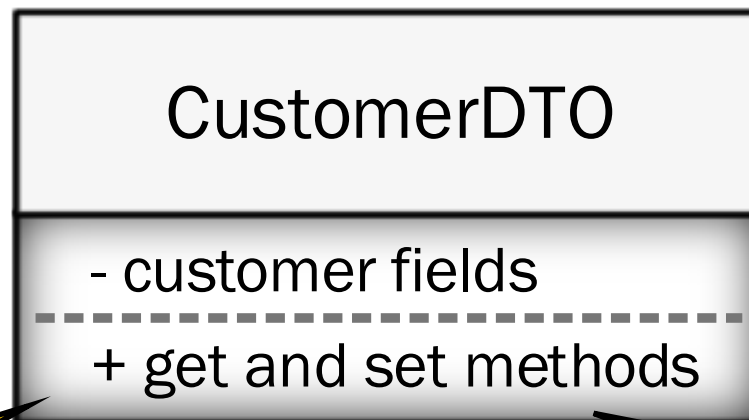


DTO: motivation

- Collecting data from a data source (e.g. database, server) is a time-consuming operation
 - The major cost is related to the round-trip time between the client and the data source
 - The **Data Transfer Object** (DTO) is one way to reduce the number of data collection calls
 - In one data collection operation, the DTO can aggregate the data that would have been collected by several operations
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DTO: in practice

- A **Data Transfer Object** (DTO) is used to carry data from one part of the application to another



A DTO has very little behaviour (e.g. serialisation and deserialisation of its own data)

DTOs are simple objects that do not contain any business logic

DTOs may be (should be?) immutable
i.e. Have no mutator (set) methods

Activity



Coding

- Review the example code and watch out for:
 - how the `Book` and `Member` classes in the project with patterns are, in essence, DTOs
 - why the `Loan` class in the project with patterns is not a DTO
- If you have any questions, ask

Project: `LectureExample_WithPatterns`